



New issue



[Web] Previously used .wasm binaries missing in 1.19.0 (using web bundler with CopyPlugin) #21811

Closed

Labels

ep:WebGPU

platform:web



e-simpson opened on Aug 21, 2024



Describe the issue

Using a onnxruntime-web with a web bundler like CopyPlugin I am now getting Package path ./webgpu is not exported from package . When I manually try to bundle the .wasm's I get Unable to locate '..node_modules/onnxruntime-web/dist/ort-wasm.wasm' glob .

I assume the names changed with this update as I see the web bundling changed. I see now bundling was introduced with [dd805ff](#) . I looked for some migration docs but couldn't find any.

To reproduce

Can follow the current quick-start https://github.com/microsoft/onnxruntime-inference-examples/tree/main/js/quick-start_onnxruntime-web-bundler

or

1. Use next.js or similar web bundler/pack framework
2. Use webpack copy plugin to load onnxruntime-web. Example for next.js:

```
/** @type {import('next').NextConfig} */
import CopyPlugin from "copy-webpack-plugin";

const nextConfig = {
  webpack: (config) => {
    config.resolve.extensions.push(".ts", ".tsx");
    config.resolve.fallback = { fs: false };

    config.plugins.push(
      new CopyPlugin({
        patterns: [
          {
            from: './node_modules/onnxruntime-web/dist/ort-wasm.wasm',
            to: 'static/chunks/pages',
```



```

    },
    {
      from: './node_modules/onnxruntime-web/dist/ort-wasm-simd.wasm',
      to: 'static/chunks/pages',
    },
    {
      from: './node_modules/onnxruntime-web/dist/ort-wasm-simd.jsep.wasm',
      to: 'static/chunks/pages',
    },
  ],
}),
);

return config;
}
}

export default nextConfig;

```

3. Import onnxruntime-web/webgpu with `import * as ort from 'onnxruntime-web/webgpu';`

4. Run the dev server, example: `bun run dev`

I get the error:

```

Module not found: Package path ./webgpu is not exported from package
.../node_modules/onnxruntime-web

```

Urgency

Not urgent, rolled back to 1.18.0 for now.

ONNX Runtime Installation

Released Package

ONNX Runtime Version or Commit ID

1.19.0

Execution Provider

'webgpu' (WebGPU)



  e-simpson added platform:web on Aug 21, 2024

  github-actions added ep:WebGPU on Aug 21, 2024



gotomypc on Aug 27, 2024



The solution to fix this, can be npm install onnxruntime-web@1.14.0, all fix the package.json, lock the onnxruntime-web to 1.14.0 version.



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e-simpson on Aug 27, 2024

Author



@gotomypc I use 1.18.0 currently which works great. I was just looking to use some new features from 1.19.0 😊



fs-eire on Aug 29, 2024

Contributor



There are multiple changes for onnxruntime-web exports.

a TL;DR version:

- if you use WebGPU, then
 - import * as ort from 'onnxruntime-web/webgpu';
 - copy ort-wasm-simd-threaded.jsep.wasm
- if you use CPU only, then
 - import * as ort from 'onnxruntime-web';
 - copy ort-wasm-simd-threaded.wasm

Just 2 .wasm files. copy the one you uses.

See [this link](#) to check out options for importing.

Let me know if you have further questions



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fs-eire on Aug 29, 2024

Contributor



work item: examples should be updated.



e-simpson on Aug 31, 2024

Author



@fs-eire thanks for the reply!

I copied the `ort-wasm-simd-threaded.jsep.wasm` into my static files, made sure I'm using `1.19.0` and importing with `import * as ort from 'onnxruntime-web/webgpu'`; I get the error:

```
Module not found: Package path ./webgpu is not exported from package
.../node_modules/onnxruntime-web (see exports field in .../node_modules/onnxruntime-
web/package.json)
> 1 | import * as ort from 'onnxruntime-web/webgpu';
    | ^
    2 |
```



Strange. I tried to copy the `ort-wasm-simd-threaded.wasm` and use the base non-gpu version and nothing seems to be working. If I revert back to `1.18.0` everything seems to work.



fs-eire on Aug 31, 2024

Contributor



[@fs-eire](#) thanks for the reply!

I copied the `ort-wasm-simd-threaded.jsep.wasm` into my static files, made sure I'm using `1.19.0` and importing with `import * as ort from 'onnxruntime-web/webgpu'`; I get the error:

```
Module not found: Package path ./webgpu is not exported from package
.../node_modules/onnxruntime-web (see exports field in
.../node_modules/onnxruntime-web/package.json)
> 1 | import * as ort from 'onnxruntime-web/webgpu';
    | ^
    2 |
```



Strange. I tried to copy the `ort-wasm-simd-threaded.wasm` and use the base non-gpu version and nothing seems to be working. If I revert back to `1.18.0` everything seems to work.

What bundler (and version) do you use? It looks like that your bundler is not reading from the "export" field of `onnxruntime-web`'s `package.json`



e-simpson on Aug 31, 2024

Author



[@fs-eire](#)

Bundling with webpack 5.93.0 (what the latest Next.js comes with)

Yeah, and I see the `1.19.0` package exports for `web-gpu`

```
"./webgpu": {
  "node": null,
  "import": "./dist/ort.webgpu.bundle.min.mjs",
  "require": "./dist/ort.webgpu.min.js",
```



```
"types": "./types.d.ts"
},
```

For reference the web-gpu exports for 1.18.0

```
"/webgpu": {
  "import": "./dist/esm/ort.webgpu.min.js",
  "require": "./dist/cjs/ort.webgpu.min.js",
  "types": "./types.d.ts",
  "default": "./dist/ort.webgpu.min.js"
},
```



Let me give turbopack a try



fs-eire on Aug 31, 2024

Contributor



Since it's Next.js, make sure not using SSR (because importing 'onnxruntime-web/webgpu' from Node.js is explicitly disabled)



illbexyz on Sep 17, 2024



I'm having the same issue. I made sure to load `onnxruntime-web/webgpu` dynamically to filter out SSR errors, but the error seems to be triggered at build time.

I made a reproduction here: <https://github.com/illbexyz/nextjs-onnxruntime>.

See <https://github.com/illbexyz/nextjs-onnxruntime/blob/main/src/pages/index.tsx> where I import `<OnnxButton />`

(which contains the import to `onnxruntime-web/webgpu`) dynamically with `ssr: false`.



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fs-eire on Sep 18, 2024

Contributor



I'm having the same issue. I made sure to load `onnxruntime-web/webgpu` dynamically to filter out SSR errors, but the error seems to be triggered at build time.

I made a reproduction here: <https://github.com/illbexyz/nextjs-onnxruntime>. See <https://github.com/illbexyz/nextjs-onnxruntime/blob/main/src/pages/index.tsx> where I import `<OnnxButton />` (which contains the import to `onnxruntime-web/webgpu`) dynamically with `ssr: false`.

I cloned the repo locally and I can reproduce this error. I am not sure how Next.js deals with this dynamic loading. But I have confirmed that even with `ssr: false`, when Next.js parses the line:

```
import * as onnx from "onnxruntime-web";
```



in `OnnxButton.tsx`, it uses the "node" export.

This is confirmed by replace the following in `node_modules/onnxruntime-web/package.json`:

```
"exports": {
  ".": {
    -   "node": {
    -     "import": "./dist/ort.node.min.mjs",
    -     "require": "./dist/ort.node.min.js"
    -   },
  +   "node": null,
    "import": "./dist/ort.bundle.min.mjs",
    "require": "./dist/ort.min.js",
    "types": "./types.d.ts"
  },
```



This will generate the following error message:

```
Module not found: Package path . is not exported from package C:\code\nextjs-
onnxruntime\node_modules\onnxruntime-web
```



Conclusion:

The error is because Next.js module resolver picks export for "node" not "import"/"require"(web) in your repo. I don't know exactly how to change this behavior, but if Next.js picks the "import"/"require"(web) export, the issue should be fixed.



Systemcluster on Sep 18, 2024



I made a reproduction here: [illbexyz/nextjs-onnxruntime](https://github.com/illbexyz/nextjs-onnxruntime).

It's missing `onnxruntime-web` in the dependencies, and both `next dev` and `next build` work for me after adding it.



fs-eire on Sep 19, 2024

Contributor



I made a reproduction here: [illbexyz/nextjs-onnxruntime](https://github.com/illbexyz/nextjs-onnxruntime).

It's missing `onnxruntime-web` in the dependencies, and both `next dev` and `next build` work for me after adding it.

Yes, you are right. It's missing in the dependencies and need to run `npm install onnxruntime-web@dev`. But I think [@illbexyz](#) is complaining about cannot import from `"onnxruntime-web/webgpu"` (while the repo uses `"onnxruntime-web"` and it works, modifying the line in `OnnxButton.tsx` will show that error)



illbexyz on Sep 19, 2024



Yes sorry my bad, I provided a quite sloppy reproduction 🙄 In case it might be useful to someone, I updated the reproduction to:

- declare `onnxruntime-web@dev` in `package.json`
- use `onnxruntime-web/webgpu` in `OnnxButton.tsx`

The error is because Next.js module resolver picks export for `"node"` not `"import"/"require"` (web) in your repo. I don't know exactly how to change this behavior, but if Next.js picks the `"import"/"require"`(web) export, the issue should be fixed.

Yes I agree with this conclusion, but I have no idea on how to configure `next.js` to pick the `"import"/"require"`(web) export.



e-simpson on Sep 20, 2024

Author



Yeah, there has to be a `"next.js"` way to get around this either with some aliases, dynamic resolving, or client flags.

However, for now, I've just worked around it and have been using `1.19.x` just as I did before in `next.js` by manually removing the `"node": null`, lines from module's `package.json`. My `onnxruntime-web/package.json` now looks like:

```
...
"exports": {
  ".": {
    "node": {
      "import": "./dist/ort.node.min.mjs",
      "require": "./dist/ort.node.min.js"
    },
    "import": "./dist/ort.bundle.min.mjs",
    "require": "./dist/ort.min.js",
    "types": "./types.d.ts"
  },
  "./all": {
    "import": "./dist/ort.all.bundle.min.mjs",
    "require": "./dist/ort.all.min.js",
```



```

    "types": "./types.d.ts"
  },
  "./wasm": {
    "import": "./dist/ort.wasm.min.mjs",
    "require": "./dist/ort.wasm.min.js",
    "types": "./types.d.ts"
  },
  "./webgl": {
    "import": "./dist/ort.webgl.min.mjs",
    "require": "./dist/ort.webgl.min.js",
    "types": "./types.d.ts"
  },
  "./webgpu": {
    "import": "./dist/ort.webgpu.bundle.min.mjs",
    "require": "./dist/ort.webgpu.min.js",
    "types": "./types.d.ts"
  },
  "./training": {
    "import": "./dist/ort.training.wasm.min.mjs",
    "require": "./dist/ort.training.wasm.min.js",
    "types": "./types.d.ts"
  }
},
...

```

And everything works as before.



fs-eire on Sep 20, 2024

Contributor



the original idea of adding "node": null, is to avoid misuse in Node.js. It seems by removing them indeed helps to use onnxruntime-web in Next.js... if not using web worker. ([#22113](#) is the issue of next step if trying to enable web worker)



👍 3



fs-eire mentioned this on Sep 21, 2024



[Next.js does not pick the correct export of dependency vercel/next.js#70296](#)



fs-eire on Sep 23, 2024

Contributor



[@illbexyz](#) I verified the solution mentioned in [vercel/next.js#70296 \(comment\)](#) works.

Fix: [fs-eire/nextjs-onnxruntime@ b7adf6f](#)

- use 'use client'; at the top of the CSR component
- use dynamic import and use if (typeof windows !== 'undefined') to wrap the dynamic import so that it will not be compiled as Node.js usage.



illbexyz on Sep 24, 2024



[@fs-eire](#) Thank you! I can confirm that the solution works correctly on Next 14. Unfortunately it doesn't work on Next <= 12 where 'use client' wasn't introduced yet, but imho it's not a big deal.

If [@e-simpson](#) agrees I think we can close the issue 😊



e-simpson on Sep 24, 2024

Author



Yes let's consider this solved 🎉



e-simpson closed this as [completed](#) on Sep 24, 2024



blueanchovy on Jan 6



for anyone using [@imgly](#) and getting this issue, nextjs 14.2.8 with @imgly/background-removal 1.5.5 works for me. trying the above solutions didn't really solve the issue for me so putting it out here



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